IHJ Cardiovascular Case Reports (CVCR) xxx (2017) xxx–xxx

Contents lists available at ScienceDirect

IHJ Cardiovascular Case Reports (CVCR)

journal homepage: www.elsevier.com/locate/ihjccr

Case report

Imaging findings in a patient with Giant Left Atrium

Kunal Mahajan*, Prakash Chand Negi, Sanjeev Asotra, Sanjay Rathour, Sachin Sondhi

Department of Cardiology, Indira Gandhi Medical College and Hospital Shimla, HP, India

ARTICLE INFO

Article history:
Received 24 May 2017
Received in revised form 4 July 2017
Accepted 23 August 2017
Available online xxx

Keywords:
Giant left atrium
Rheumatic heart disease
Echocardiography

ABSTRACT

A 65 years old female with rheumatic heart disease presented to us with worsening breathlessness and palpitations. Echocardiogram revealed a giant left atrium. Here we discuss the clinical and imaging clues to the diagnosis of giant left atrium.

© 2017 Cardiological Society of India. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Introduction

Giant LA, measuring more than 80 mm by definition, is almost always the result of rheumatic mitral valve disease, predominantly regurgitation.[1] It is believed that chronic volume and pressure overload created by mitral valve disease, in addition to weakening of left atrial wall because of rheumatic pancarditis, is responsible for the excessive enlargement of LA.[2]

Case report

A 65-years-old female with past history of rheumatic heart disease presented with complaints of worsening breathlessness and palpitations. She denied any history of exertional chest pain, giddiness or fainting, swelling of feet, hoarseness of voice, difficulty in swallowing or stroke. On examination she was in atrial fibrillation with a fast ventricular rate. Her blood pressure was 90/60 mm Hg. She had cardiomegaly and a grade 3 parasternal heave. Cardiac auscultation revealed a soft S1 along with a pansystolic murmur at apex radiating to axilla. A long mid diastolic rumbling murmur across the mitral area was also heard. Examination of other organ systems was normal. Chest X-Ray showed increased cardiothoracic ratio with splaying of carina and elevation of left main bronchus suggestive of left atrial enlargement. (Fig. 1) Transthoracic echocardiography demonstrated a giant left atrium (LA) with swirling spontaneous echo contrast in it. (Fig. 2, Video 1) How-

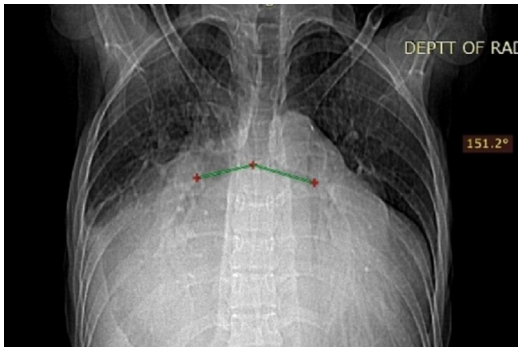


Fig. 1. Chest X-ray revealing massive cardiomegaly with left atrial enlargement characterized by widening of carinal angle (151^0) and elevation of left main bronchus. Note that the enlarged left atrium has caused opacification of right lung. Also note cephalization of upper lobe veins.

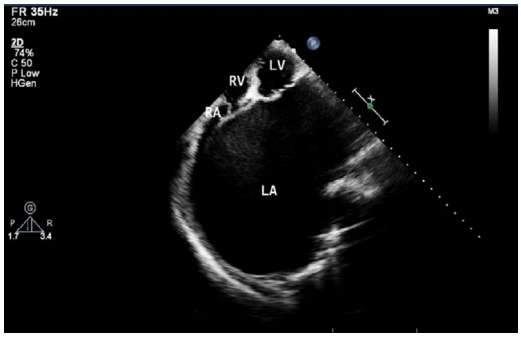


Fig. 2. Transthoracic echocardiogram-apical-4-chamber view showing a massively enlarged left atrium.

* Corresponding author at: Dr Kunal Mahajan, Senior Resident Cardiology, IGMC Shimla, 171001, India.
E-mail address: kunalmahajan442@gmail.com (K. Mahajan).

http://dx.doi.org/10.1016/j.ihjccr.2017.08.007
2468-600X/© 2017 Cardiological Society of India. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

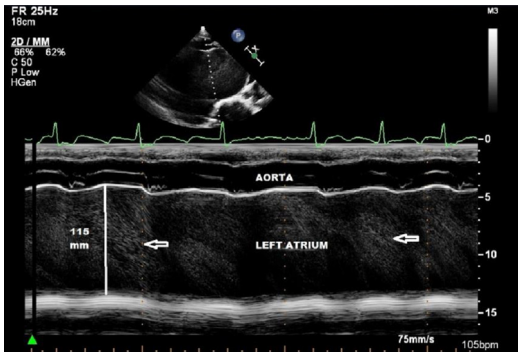


Fig. 3. Transthoracic echocardiogram- parasternal long axis view showing the enlarged left atrium with an anterior-posterior diameter of 115 mm. Also note the spontaneous echo contrast in left atrium. (arrows).

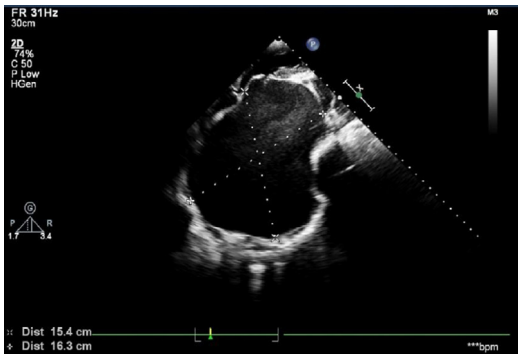


Fig. 4. Transthoracic echocardiogram-apical-4-chamber view showing an enlarged left atrium with dimensions 154 × 163 mm. Note the diastolic doming of anterior mitral leaflet.

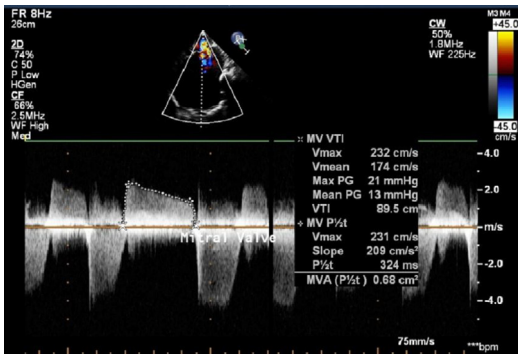


Fig. 5. Transthoracic echocardiogram showing the mitral valve area (by pressure half time) and gradients across the mitral valve.

ever, there was no definite evidence of a thrombus. LA measured 115 mm in parasternal long axis view (Fig. 3) and 154 × 163 mm in apical-4-chamber view. (Fig. 4) Mitral valve was diseased with thickened and calcified leaflets and mitral valve area was measured to be 0.6 cm2. (Fig. 5) Both the leaflets showed restricted movements with diastolic doming of anterior mitral leaflet. These findings were consistent with severe rheumatic mitral stenosis. In addition moderate eccentric mitral regurgitation (Fig. 6) and severe subvalvular deformity were also present making it unsuitable for balloon mitral valvotomy. Computerized tomography scan revealed a massively enlarged LA (193 × 133 × 167 mm). (Figs. 7–9) Its massive size made it one of the largest LA ever reported. Patient was advised mitral valve replacement. However she refused and was followed up on medical management.

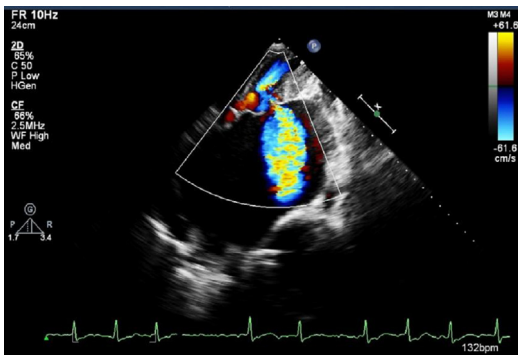


Fig. 6. Echocardiogram showing an eccentric jet of moderate mitral regurgitation.

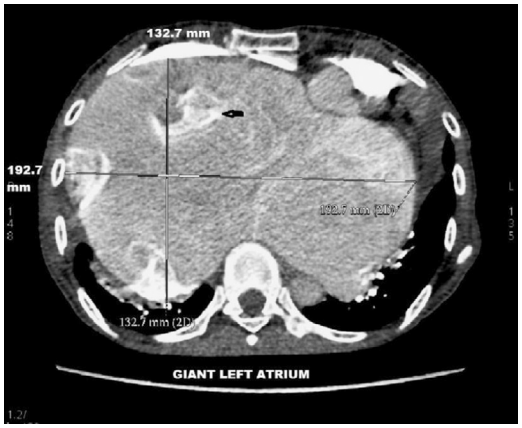


Fig. 7. Computed tomography scan-axial view, revealed massive left atrial enlargement.

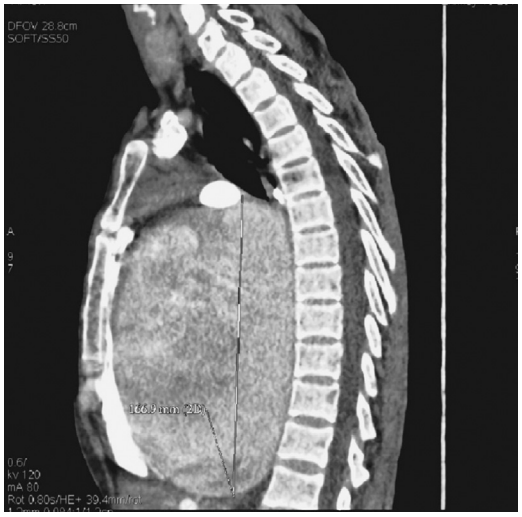


Fig. 8. Computed tomography scan- sagittal view, showing an enlarged left atrium, measuring 167 mm in the superior-inferior axis.

Discussion

Left atrial anterior-posterior diameter of greater than 80 mm on transthoracic echocardiography is considered diagnostic of giant left atrium. Giant left atrium occurs almost exclusively in patients with rheumatic mitral valve disease with mitral regurgitation being the predominant lesion.[3–6] Giant left atrium should always be suspected in a patient of rheumatic mitral disease who develops right lung opacification on chest X-ray.[2] It predisposes to increased risk

Please cite this article in press as: Mahajan K, et al, Imaging findings in a patient with Giant Left Atrium, *IHJ Cardiovasc Case Rep*. (2017), http://dx.doi.org/10.1016/j.ihjccr.2017.08.007

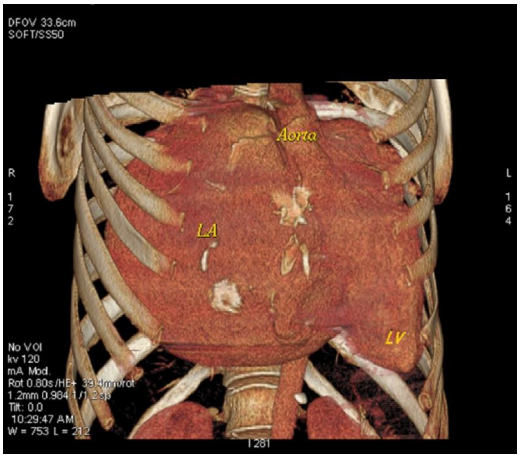


Fig. 9. Reconstructed CT image showing a hugely dilated left atrium extending to the right of aorta and occupying almost whole of the thorax.

of atrial fibrillation, thromboembolism, and compressive symptoms of dysphagia and hoarseness of voice. Surgery in giant left atrium aims at correcting the mitral valve abnormalities, relieving the compression symptoms, prevention of thromboembolism and reversion of atrial fibrillation to normal sinus rhythm. Two strategies had been applied in the management of mitral valve surgery with giant left atrium. The first is performing mitral valve surgery with left atrial volume reduction (debulking) while the second is to perform mitral valve replacement alone.[7] Mitral valve replacement alone is believed to result in reduction of left atrial size by decreasing pressure and volume overload on left atrium. However associated left atrial debulking may be required in presence of compressive symptoms. Left atrial debulking may also prevent recurrent thrombosis by reducing intra-atrial stasis.[7]

Source of funding

None.

Conflicts of interest of each author/contributor

None.

Authors contribution

All the authors were actively involved in the management of the patient and preparation of this manuscript.

Manuscript has been read and approved by all the authors, and the requirements for authorship have been met, and each author states that the manuscript represents honest work.

Acknowledgement

none.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.ihjccr.2017.08.007.

References

1. Loch A, Sadiq MA, Wan Ahmad WA. Giant left atrium in a patient with prosthetic mitral valve. *Eur Heart J*. 2013;34(13):981.
2. Hurst W. Memories of patients with a giant left atrium. *Circulation*. 2001;104:2630–2631.
3. Goldenberg G, Eisen A, Weisenberg N, Amital H. Giant left atrium. *IMAJ*. 2009;11:641.
4. Tung R, DeSanctis R. Giant left atrium. *NEJM*. 2004;351:1437.
5. Ates M, Sensoz Y, Abay G, Akcar M. Giant left atrium with rheumatic mitral stenosis. *Texas Heart Inst J*. 2006;33:389–391.
6. Rajakaruna C, Mhandu P, Ghosh-Dastidar M, Desai J. Giant left atrium secondary to severe mixed mitral valve pathology. *Eur J Cardiol Thor Surg*. 2007;32:932.
7. Apostolakis E, Shuhaiber JH. The surgical management of giant left atrium. *Eur J Cardiothorac Surg*. 2008;33:182–190.

Please cite this article in press as: Mahajan K, et al, Imaging findings in a patient with Giant Left Atrium, *IHJ Cardiovasc Case Rep*. (2017), http://dx.doi.org/10.1016/j.ihjccr.2017.08.007